SOP-PHM-005 PHM-200 pH Meter Calibration and Error Codes

DEMO DATA - written for the SmartLab AI demonstration. This is not an approved laboratory procedure.

1. Calibration
Calibrate the meter daily with pH 4.01, 7.00 and 10.01 buffers at room temperature. Rinse the electrode with deionised water between buffers and blot it dry with lint-free tissue; never wipe the glass bulb.

2. Error CAL-02 (slope out of range)
CAL-02 means the electrode slope is outside 90 to 105%. Replace the calibration buffers with fresh ones and repeat the calibration. If CAL-02 persists, clean the electrode by soaking it in 0.1 M HCl for 15 minutes, rinse, and recalibrate. Replace the electrode if the error continues.

3. Error ERR-07 (unstable reading)

ERR-07 means the reading did not stabilise within 60 seconds. Check that the electrode junction is wet and that the reference fill solution covers the internal element. Refill with 3 M KCl if the level is low. Store the electrode in storage solution, never in deionised water.

4. Safety

Wear safety glasses and gloves when handling the hydrochloric acid cleaning solution.